

The cost of a mark: a Newton-age floor for the Galileo comparison

Newton takes the sagitta and the swept area of the Galileo comparison to zero in Lemmas X–XI and keeps only their ratio to the time. That limit is a statement about the unmarked curve, and it has no floor: every partition of the fall carries impulses, sagittas and areas that vanish with the mesh (Proposition 1). A floor appears as soon as the comparison must be *recorded*, and it appears from three premises that Newton’s own optics supplies: light is made of least corpuscles of impulse p (M1), a mark made with light of one colour fixes a position no better than the interval of fits Λ (M2), and a mark that fixes a position within δ leaves the impulse it delivered undetermined within $\Lambda p/\delta$ (M3). Under M1–M3 the falling parabola can be told from the inertial line over a duration τ only if

$$\tau \Delta E = \frac{F^2 \tau^3}{2m} > 8\Lambda p, \quad A_{\text{inertial,fall}} = \frac{vF\tau^3}{6m} > \frac{8v\Lambda p}{3F},$$

and Democritus insertion of marks into the trajectory stops at the mesh $\tau_* = (16m\Lambda p/F^2)^{1/3}$: finer marks record free motion only (Theorems 1–2). Dropping M3 and keeping Newton’s determinate fits gives an exact countermodel with floor zero (Section 5), so M3 is the single premise that carries $h > 0$; it is a statement of indeterminacy, and no Newton-age mechanics or optics derives it. Newton measured $\Lambda = 1/89000$ inch; he could not measure p ; and his corpuscle sizes make Λp colour dependent. On the modern identification $\Lambda p = h/2$ the floor is $4h$. This is N02’s principle, obstruction and countermodel. Two later notes remove its two weaknesses. The derivation note removes the protocol dependence: for every finite protocol of marks with $\delta\Delta \geq \kappa$ the comparison needs $F^2\tau^3 > 9m\kappa$, and two marks suffice above $36m\kappa$. The mark-cost note identifies M3 and fixes its constant: for a mark made by coupling the body’s position to a probe, the error operator and the impulse delivered are canonically conjugate, so $\kappa \geq \hbar/2$ exactly by Robertson’s inequality, and M3 is that inequality stated in Newton’s two quantities. In the Gaussian statistical model the floor becomes $\tau\Delta E \geq 48z_{1-\epsilon}^2\kappa$, sharp, for marks whose error and recoil are uncorrelated; a correlation ρ lowers the effective cost to $\frac{\hbar}{2}\sqrt{(1-\rho)/(1+\rho)}$, which is the one resource that moves the floor. An earlier version of this paragraph reported $\kappa \geq 2\hbar$ from the derivation note’s first Proposition 6; that value was too large by four and invalidly proved, and the worst-case widths used here have no non-vacuous quantum instance at all. Exploratory; no ledger promotion.

1. What Newton’s limit commits to

Take $m, F, v, \tau > 0$, horizontal coordinate x , downward coordinate y , inertial path $q_I(t) = (vt, 0)$ and falling path $q_F(t) = (vt, Ft^2/2m)$ on $[0, \tau]$, as in N01. The sagitta, the inertial–fall area and the gained energy are

$$s(\tau) = \frac{F\tau^2}{2m}, \quad A(\tau) = \frac{vF\tau^3}{6m}, \quad \Delta E(\tau) = \frac{F^2\tau^2}{2m}, \quad \frac{3F}{v}A = \tau\Delta E.$$

Lemma X says $s \propto \tau^2$ at the start of the motion; Lemma XI, Corollaries 4–5, that the curved segment is one third of the tangent triangle and both scale as τ^3 ; Proposition VI reads the force off the limit $2ms/\tau^2$; Proposition I builds the orbit as a polygon with “a single but great impulse” at each vertex and lets the number of vertices be “augmented in infinitum” (Motte/Wilkins, passage; the Principia note has the paragraph map). The impulses, sagittas and areas at the vertices all vanish in that limit; the force survives as a ratio.

Proposition 1 (no geometric floor). For every partition $0 = t_0 < t_1 < \dots < t_n = \tau$ with steps τ_j and mesh $|\pi| = \max \tau_j$, the vertex impulse $F\tau_j$, the step sagitta $F\tau_j^2/2m$ and the step area $vF\tau_j^3/6m$ tend to zero with the mesh, their sum obeys $\sum_j vF\tau_j^3/6m \leq vF\tau|\pi|^2/6m$,

and the ratio $2ms(\tau_j)/\tau_j^2 = F$ is exact at every step. No partition, however fine, produces a contradiction.

Proof. Direct from the formulas; the bound on the sum is C027’s computation with the constant-force chord replaced by the inertial tangent. $\square$

Berkeley’s objection that evanescent increments are “neither finite Quantities nor Quantities infinitely small, nor yet nothing” (Analyst §35, passage) is answered inside the geometry by the limit ratio, as C027–C029 and the full-state classical cuts answer it inside mechanics. A floor therefore has to come from what it takes to *mark* the curve, since refinement consistency of the curve offers none. That is the reading of “Democritus insertion” used here: inserting a point into the trajectory means making a physical mark on the body at a time, and marks have a cost.

2. The comparison as a record

A **mark** is an interaction between the body and a recording system that leaves a datum about the transverse coordinate y at a time t . Its **resolution** δ is the width of the interval to which the datum confines $y(t)$. Its **recoil width** Δ is the width of the interval to which the totality of records, made before or after the mark, confines the transverse impulse the mark delivered to the body. Both are worst-case widths; nothing probabilistic is assumed.

Two-mark protocol. A preparation mark at $t = 0$ with (δ_0, Δ_0) and a terminal mark at $t = \tau$ with resolution δ_1 . From the preparation the inertial prediction is an interval I_I of width

$$w = \delta_0 + \frac{\Delta_0 \tau}{m},$$

and the falling prediction is $I_F = I_I + s(\tau)$. The terminal datum decides between them in the worst case exactly when

$$s(\tau) > w + \delta_1. \tag{1}$$

The velocity v plays no role in (1); it enters only the conversion of the action-scaled product $\tau \Delta E$ into the geometric area A .

Insertion protocol. Marks at $t_j = j\tau/n$, each with (δ_j, Δ_j) . The force is *recorded on step* j when the comparison restricted to $[t_j, t_{j+1}]$, with mark j as preparation and mark $j + 1$ as terminal, satisfies (1) with τ replaced by $\tau_n = \tau/n$.

3. Three premises from Newton’s optics

All four Opticks passages below are stored with their line anchors in the source companion, which prints them verbatim from the 1730 fourth edition.

- **(M1) Least impulse.** Light consists of corpuscles (“Are not the Rays of Light very small Bodies emitted from shining Substances?”, Opticks Query 29, passage). Corpuscles of one colour are alike; each carries an impulse $p > 0$, and no part of the light of that colour carries less. This is Democritus’ least part applied to the probe.
- **(M2) Least length.** A mark made with light of one colour has resolution $\delta \geq \Lambda$, the interval of fits of that colour. Newton defines the interval as “the space it passes between every return and the next return” of the disposition to be reflected (Book II Part III, Definition after Prop. XII, passage), and measures it: “the Intervals of their Fits of easy Reflexion are the 1/89000th part of an Inch” for the yellow–orange confine (Prop. XVIII, passage), that is $\Lambda \approx 0.285 \mu\text{m}$. A datum obtained through the fits repeats every Λ and so fixes a length no better than Λ ; Newton’s rings are exactly such a gauge.
- **(M3) Mark trade-off.** A mark made with light of one colour that fixes the position within δ has recoil width $\Delta \geq \Lambda p / \delta$. Newton’s theory motivates the *pre-mark* half of this: whether a corpuscle is reflected or transmitted at the body is settled by its fit, the fit

depends on the path length modulo Λ , and that path length is the very position the mark is to find, so no record made before the mark fixes the delivered impulse ($2p$ on reflection, about 0 on transmission) better than p ; confining the ray to a breadth δ adds the inflexion of Book III (“Do not Bodies act upon Light at a distance, and by their action bend its Rays”, Query 1, passage; Grimaldi 1665, metadata) with angles of order Λ/δ . The premise asserts more: that records made *after* the mark do not recover the impulse either. Section 5 shows this is the whole content.

The dimensional gate of Q14, Theorem A, sorts the premises before any calculation. The marking model’s fixed constants are Λ , with dimension vector $(0, 1, 0)$, and p , with $(1, 1, -1)$. Their unique action product is Λp . A model with M2 alone (fixed Λ) or M1 alone (fixed p) holds no action product and so has floor 0 or ∞ for every observable; Section 5 shows it is 0. Both least quantities are needed, and they are needed as a product, which is what M3 says. The speed of light c does not enter the protocol; admitting it would only add the mass p/c and a possible factor $F(mc/p)$ that the argument below never produces.

4. The floor and the insertion mesh

Theorem 1 (floor for the Galileo comparison). Under M1–M3, with the preparation and terminal marks made by light of one colour, the two-mark protocol decides the comparison in the worst case only if

$$s(\tau) > 2\sqrt{\frac{\Lambda p \tau}{m}} + \delta_1, \quad \text{hence} \quad \tau \Delta E = \frac{F^2 \tau^3}{2m} > 8\Lambda p, \quad A > \frac{8v\Lambda p}{3F}.$$

Proof. By M3, $\Delta_0 \geq \Lambda p/\delta_0$, so $w \geq \delta_0 + \Lambda p \tau/(m\delta_0) \geq 2\sqrt{\Lambda p \tau/m}$, with equality at $\delta_0 = \sqrt{\Lambda p \tau/m}$. Insert in (1) and drop $\delta_1 \geq \Lambda > 0$: $F\tau^2/2m > 2\sqrt{\Lambda p \tau/m}$. Squaring, $F^2\tau^4/4m^2 > 4\Lambda p \tau/m$, so $F^2\tau^3 > 16m\Lambda p$, which is the displayed bound; the area follows from $(3F/v)A = \tau \Delta E$. $\square$

The constant is protocol dependent, the form is not. A three-mark protocol with no prepared velocity, marks at $0, \tau/2, \tau$ and the mid-chord test $y(\tau/2) - \frac{1}{2}[y(0) + y(\tau)]$, has deviation $s(\tau)/4$; the first recoil cancels in the chord test, the second enters as $\Delta\tau/4m$, the three resolutions enter with weights $1, \frac{1}{2}, \frac{1}{2}$, and the same minimization gives $F^2\tau^3 > 128m\Lambda p$, that is $\tau \Delta E > 64\Lambda p$. Any protocol built from marks obeying M1–M3 gives $\tau \Delta E > \kappa \Lambda p$ with a positive number κ fixed by the protocol, because (Theorem A) Λp is the only action the model contains. The action-scaled product carries the universal floor; the geometric area carries the extra factor v/F . This is the Q14 gate seen from Newton’s side, and the reason the anchor’s conversion factor $3F/v$ is indispensable.

Theorem 2 (insertion mesh). Under M1–M3, in the insertion protocol the force is recorded on a step only if $\tau_n > \tau_*$, where

$$\tau_* = \left(\frac{16m\Lambda p}{F^2} \right)^{1/3}, \quad n < n_* = \frac{\tau}{\tau_*} = \tau \left(\frac{F^2}{16m\Lambda p} \right)^{1/3},$$

and at the finest recording mesh each step’s area is $A(\tau_*) = 8v\Lambda p/3F$. Over k consecutive steps of a finer mesh the comparison becomes decidable exactly when $k\tau_n > \tau_*$.

Proof. Theorem 1 applied to one step of duration τ_n , then to the duration $k\tau_n$, using the marks at the ends and ignoring the interior marks, which the observer may make with $\delta \rightarrow \infty$ and hence zero recoil. $\square$

Below τ_* every recorded step is consistent with inertial motion. The polygon of Proposition I can be refined past n_* , and its vertices still carry Newton’s impulses $F\tau_n$ in the geometry, but no mark shows them: the body is free at every recorded instant and curved over any $k > n/n_*$ of them. Zeno’s arrow and Democritus’ cone (I003, remark 1) receive one answer: adjacent

recorded instants are equal, adjacent recorded slices are equal, and the difference lives in the unrecorded accumulation over more than n/n_* of them.

5. The countermodel: determinate fits

Keep M1 and M2 and take the fits and the inflexion as Newton took them, as determinate dispositions: the corpuscle’s fate at the body is a function of its phase and its distance from the edge, and the corpuscle then travels freely to a screen at distance D , where a second mark of resolution $\delta_s \geq \Lambda$ records its landing. The corpuscle’s transverse momentum after the mark is then fixed by the two positions to within $p(\delta + \delta_s)/D$, and by momentum conservation so is the impulse it left in the body. Since D is free, the recoil width $\Delta \rightarrow 0$ at fixed $\delta \geq \Lambda$: M3 fails. The two-mark protocol then needs only $s(\tau) > 2\Lambda + \delta_1$, roughly $F\tau^2 > 6m\Lambda$, and

$$\tau\Delta E > \frac{F^2}{2m} \left(\frac{6m\Lambda}{F} \right)^{3/2} = \sqrt{54 m F \Lambda^3} \rightarrow 0 \quad (F \rightarrow 0),$$

so the floor over admitted forces and masses is zero, as Theorem A(i) requires of a model whose only fixed constant is a length. This is the same closure the classical apparatus results record: position and momentum records with independent precisions close the canonical product (C068–C123, synthesis), and the force lens of C070–C071, whose canonical area $2F^2\ell^3/3m$ is the same combination as $\tau\Delta E$, closes with the delay. Theorem 1 is the statement that this lens cannot be resolved below a fixed multiple of Λp once M3 holds.

The countermodel isolates the premise. Ignorance of the fit before the mark, which Newton’s theory grants, is removed by the far screen and is too weak. M3 therefore means that no record, before or after, fixes both where the corpuscle struck within δ and what impulse it left within $\Lambda p/\delta$: the delivered impulse is not a function of the records. In later language this is the complementarity of the probe’s position and momentum records. A Newton-age physicist can state M3, can see from the fits that it is at least true of all prior records, and cannot derive it; What Newton printed entails the negation (a determinate disposition carried by a corpuscle under deterministic mechanics; he states neither M3 nor its negation). Everything in Theorems 1–2 follows from M3 with Newton’s other materials, and nothing in them follows without it. This is the precise sense in which the Newton-age argument for $h > 0$ exists: it is one premise long, and the premise is the one Newton’s determinism excludes.

6. Newton’s numbers, universality and the quantum value

Newton had Λ to three figures and had no way to measure p , so his floor $\kappa\Lambda p$ is positive and unknown. He also had the colour dependence of Λ : the ring intervals for the utmost red and utmost violet are “as 14 to 9” (Book II Part I, Obs. 13, passage), and Query 29 makes the red corpuscles the biggest and the violet the least (passage). If bigger means more impulse, Λp increases toward the red and the floor over the spectrum is its violet value; if size and impulse are unrelated the floor is $\min_{\text{colour}} \Lambda p$, positive because the spectrum is bounded. In Q14’s terms the fixed constants $\{\Lambda_i, p_i\}$ give the action products $\Lambda_i p_j$ and the dimensionless ratios Λ_i/Λ_j , p_i/p_j , so Theorem A(ii) allows any positive function of them; Newton-age data fix its positivity and not its value or constancy.

Universality across colours is what Planck’s constant adds and what the Newton-age argument cannot reach. With $p = h/\lambda$ and $\Lambda = \lambda/2$ (the fits alternate every half wavelength), $\Lambda p = h/2$ for every colour, and Theorem 1 reads $\tau\Delta E > 4h$. The quantum benchmark of N01 is the same combination with a different criterion: perfect discrimination of the closed-loop phase needs $|\Delta S_{\text{loop}}| = \frac{4}{3}\tau\Delta E \geq \pi\hbar$, that is $\tau\Delta E \geq 3h/8$. The factor between $4h$ and $3h/8$ is the difference between worst-case intervals and optimal quantum discrimination, and between an open two-mark record and a closed interferometric one. Both say that the inertial-fall lens must carry an action-scaled product of order h before it is a distinguishable alternative.

7. Where $h \rightarrow 0$ and $h > 0$ both hold

Newton’s limit lives in the geometry of the unmarked curve, where Proposition 1 shows it is sound and C027 shows the chord error closes as $|\pi|^2$ at every mesh, marks or no marks. The floor lives in the marks. The two are compatible because the force is a ratio approached from durations above τ_* , where the comparison is recordable, and because Lemma I’s “ultimately equal” quantities become literally equal for every record once the mesh passes τ_* . Below that mesh Berkeley’s departed quantities are the sagitta and the interval of a step that no mark can distinguish from free motion, and their ratio is still F because the geometry does not depend on being marked.

For I003 this places the two remarks. Remark 2, the obstruction to mapping time to positions as a double limit, is Theorem 1 and needs M3. Remark 3, an h that controls the convergence, is Theorem 2 and the mesh τ_* : the marked polygon converges to Newton’s curve on meshes above τ_* and stops carrying information below it. The area-rate floor $GM/2c$ noted earlier in STATE is a rate and gives no area floor, since $GM\tau/2c \rightarrow 0$; it is set aside.

8. Consequence for STATE

N02 asked for one non-quantum consistency principle for joint time/position refinement of the Galileo comparison, its cut state and composition, and either a zero-action contradiction or an exact countermodel. The principle is M3, stated for marks; the cut state at a mark is the pair of intervals (δ, Δ) ; composition is the worst-case propagation of Section 2; the contradiction with zero-action refinement is Theorems 1–2; the countermodel is Section 5. The principle turns out to be an added indeterminacy premise, statable in Newton’s terms and excluded by what he printed, and it is the entire difference between a floor and none; Newtonian mechanics and Newton’s optics contain no consistency requirement that yields it. The user’s necessity question is thereby sharpened to a question about M3 alone.

Next: (a) ask whether M3 has a consistency derivation of its own, for instance from requiring that marks on probes obey the same premise as marks on bodies without a far-screen loophole, which is the point where a non-commuting record structure must enter (G07’s “two scale-free non-commuting structures”); (b) state Theorem 1 for a general force law and for the C070–C071 lens, where the same $F^2\tau^3/m$ combination already appears, so that the floor is a phase-space statement rather than a parabola statement; (c) done: the Opticks companion now holds Props. XII and XVIII, the Definition of the interval, Obs. 13 and Queries 1 and 29 verbatim with line anchors.